Math 141, S10 **Name:** Answer

Quiz 7 **Section:** **SSN:** xxx-xx-

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(5 points) Use **implicit differentiation** to find dy/dx of

$$x^3 + 2y^3 = 1.$$

Solution: $\frac{d}{dx}[x^3 + 2y^3] = \frac{d}{dx}[1]$

$$\frac{d}{dx}[x^3] + \frac{d}{dx}[2y^3] = 0$$

$$3x^2 + 6y^2\frac{dy}{dx} = 0$$

$$6y^2\frac{dy}{dx} = -3x^2$$

$$\Rightarrow \frac{dy}{dx} = -\frac{3x^2}{6y^2} \Rightarrow \frac{dy}{dx} = -\frac{x^2}{2y^2}$$

Problem 2.(5 points) Use **logrithmic differentiation** to find the derivative of

$$y = x^{\sin x}.$$

Solution: First take logarithm on both sides, we get

$$\ln y = \ln(x^{\sin x})$$

$$= \sin x \ln x$$

Then $\frac{d}{dx}[\ln y] = \frac{d}{dx}[\sin x \ln x]$

$$\frac{y'}{y} = \sin x \cdot \left(\frac{1}{x}\right) + (\ln x) \cdot \cos x$$

$$y' = y\left(\frac{\sin x}{x} + \cos x \ln x\right)$$

$$= x^{\sin x}\left(\frac{\sin x}{x} + \cos x \ln x\right)$$